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\\Research\Investigators\Frederick\Keto MRS 20240709\localizer
TA: 19 sec Coil Selection: Auto Voxel Size: 0.5×0.5×10.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
TR	20.0 ms
TE	5.00 ms
Averages	1
Concatenations	7
AutoAlign	---

Contrast - Common

TR	20.0 ms
TE	5.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	40 deg

Contrast - Common

Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
Base Resolution	256
Phase Resolution	50 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal

Geometry - Common

Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
TR	20.0 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	7

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	L >> R
Coronal	P >> A
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard

System - Adjustments

B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	20.0 ms
Segments	1
Concatenations	7

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	280 mm
FOV Phase	100.0 %
Phase Resolution	50 %

Physio - PACE

Resp. Control	Off
Concatenations	7

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	180 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\Investigators\Frederick\Keto MRS 20240709\AAHScout
TA: 17 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	All Segments
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.15 ms
TE	1.37 ms
Averages	1
Concatenations	1
AutoAlign	Head

Contrast - Common

TR	3.15 ms
TE	1.37 ms
Flip Angle	8 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	7.5 s

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.15 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm

Geometry - AutoAlign

H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\Keto MRS 20240709\T1_MEMPRAGE_64ch

TA: 6:02 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R2.2 P1.7 F31.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2530.0 ms
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain
Coil Elements	HC1-7

Contrast - Common

TR	2530.0 ms
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
Magn. Preparation	Non-sel. IR
TI	1100 ms
Flip Angle	7 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	4
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R2.2 P1.7 F31.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2530.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R2.2 P1.7 F31.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R2.2 P1.7 F31.8
R	2.2 mm
P	1.7 mm
F	31.8 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	32 mm
Table Position	F
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R2.2 P1.7 F31.8 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249403 MHz

System - Tx/Rx

? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2530.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	Non-sel. IR
TI	1100 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	tfl_me
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Reordering	Linear

Sequence - Part 1

Bandwidth 1	650 Hz/Px
Bandwidth 2	650 Hz/Px
Bandwidth 3	650 Hz/Px
Bandwidth 4	650 Hz/Px
Echo Spacing	9.84 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Readout trajectory	Bipolar
Gradient spoiling	Siemens
Gradient moment factor	1.00
Averaging	RMS

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\Keto MRS 20240709\t2_tse_dark-fluid_tra
TA: 4:14 min Coil Selection: Auto Voxel Size: 0.7×0.7×4.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	33
Distance Factor	25 %
Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	96.9 %
Slice Thickness	4.0 mm
TR	9000.0 ms
TE	83.00 ms
Averages	1
Concatenations	2
AutoAlign	Head > Brain

Contrast - Common

TR	9000.0 ms
TE	83.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	Slice-sel. IR
TI	2500 ms
Freeze Suppr. Tissue	On
Flip Angle Mode	Constant
Flip Angle	150 deg
Fat-Water Contrast	Fat Saturation
Fat Saturation	Strong
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard

Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	220 mm
FOV Phase	96.9 %
Slice Thickness	4.0 mm
Base Resolution	320
Phase Resolution	70 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	33
Distance Factor	25 %
Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	96.9 %
Slice Thickness	4.0 mm
TR	9000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	2

Geometry - AutoAlign

Slice Group	1
Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L1.6 A6.8 F2.6
L	1.6 mm
A	6.8 mm
F	2.6 mm
Initial Orientation	T > C

Geometry - AutoAlign

T > C	-9.80
> S	0.00
Initial Rotation	89.56 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Rotation	90.90 deg
R >> L	214 mm
A >> P	220 mm
F >> H	164 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	9000.0 ms
Concatenations	2

Physio - Cardiac

Fat-Water Contrast	Fat Saturation
Magn. Preparation	Slice-sel. IR
TI	2500 ms
Dark Blood	Off
FOV Read	220 mm
FOV Phase	96.9 %
Phase Resolution	70 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	2

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Motion Correction	None
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	tir
Dimension	2D
RF Pulse Type	Normal
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	289 Hz/Px
Echo Spacing	8.28 ms
Free Echo Spacing	Off
Define	Turbo Factor
Turbo Factor	17
Echo Trains per Slice	13

Sequence - Part 2

Introduction	On

Sequence - Part 2

Phase Correction	Automatic
Compensate T2 Decay	Off
Fast Mode	Off
WARP	Off
Red. EC Sensitivity	Off
Acoustic noise reduction	Off
Reduce Motion Sens.	On
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	60 s

\\Research\Investigators\Frederick\Keto MRS 20240709\pd+t2_tse_tra
TA: 2:11 min Coil Selection: Auto Voxel Size: 0.3×0.3×4.0 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	33
Distance Factor	25 %
Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	3600.0 ms
TE 1	9.40 ms
TE 2	94.00 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	3600.0 ms
TE 1	9.40 ms
TE 2	94.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	160 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	2
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
Base Resolution	320
Phase Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	30
Deep Resolve	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	33
Distance Factor	25 %
Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	3600.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L1.6 A6.8 F2.6
L	1.6 mm
A	6.8 mm
F	2.6 mm
Initial Orientation	T > C
T > C	-9.80

Geometry - AutoAlign

> S	0.00
Initial Rotation	89.56 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3600.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	100.0 %
Phase Resolution	100 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Motion Correction	None
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	tse
Dimension	2D
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	252 Hz/Px
Echo Spacing	9.36 ms
Free Echo Spacing	Off
Define	Turbo Factor
Turbo Factor	5
Echo Trains per Slice	35

Sequence - Part 2

Introduction	On
Phase Correction	Automatic

Sequence - Part 2

Compensate T2 Decay	Off
Fast Mode	Off
WARP	Off
Red. EC Sensitivity	Off
Acoustic noise reduction	Off
Reduce Motion Sens.	On
Motion Correction	None

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	130 deg
Allowed Delay	60 s

\\Research\Investigators\Frederick\Keto MRS 20240709\resolve_4scan_trace_tra_p2_192

TA: 1:55 min Coil Selection: Auto Voxel Size: 1.1×1.1×4.0 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	30
Distance Factor	25 %
Position	L1.6 A6.8 F2.6 mm
Orientation	T > C-9.8
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	4190.0 ms
TE 1	54 ms
TE 2	92 ms
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	4190.0 ms
TE 1	54 ms
TE 2	92 ms
MTC	Off
Magn. Preparation	None
Flip Angle	180 deg
Fat-Water Contrast	Fat Saturation
Fat Saturation	Strong
Contrasts	2
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Resolution - Common

FOV Read	220 mm

Resolution - Common

FOV Phase	100.0 %
Slice Thickness	4.0 mm
Base Resolution	192
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Accel. Mode	GRAPPA
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	96
Phase Partial Fourier	Off
Readout Partial Fourier	5/8
Readout Segments	7

Resolution - Filter

Raw Filter	On
Distortion Correction	2D
Normalize	Prescan
Noise Masking	Off

Geometry - Common

Slice Group	1
Slices	30
Distance Factor	25 %
Position	L1.6 A6.8 F2.6 mm
Orientation	T > C-9.8
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	4190.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L1.6 A6.8 F2.6 mm
Orientation	T > C-9.8
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L1.6 A6.8 F2.6
L	1.6 mm
A	6.8 mm
F	2.6 mm
Initial Orientation	T > C
T > C	-9.80
> S	0.00

Geometry - AutoAlign

Initial Rotation	-0.44 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	3 mm
Table Position	F
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L1.6 A6.8 F2.6 mm
Orientation	T > C-9.8
Rotation	-0.44 deg
A >> P	220 mm
R >> L	220 mm
F >> H	149 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	4190.0 ms
Concatenations	1

Diff

Diffusion Mode	4-Scan Trace
Diff. Directions	4
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm²
b-value 2	1000 s/mm²
Averages 1	1
Averages 2	1
Invert Gray Scale	Off
Diff. Weighted Images	Off
Trace Weighted Images	On
Tensor	Off
FA Maps	Off
ADC Maps	On
Exponential ADC Maps	Off
b-value >=	0 s/mm²
ADC Noise Threshold	100
Noise Masking	Off
Calculated Image	Off

Sequence - Part 1

Sequence Name	resolve
Dimension	2D
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	766 Hz/Px
Echo Spacing	0.32 ms
Optimization	Min. TE
EPI Factor	96

Sequence - Part 2

Introduction	On
Reacquisition Mode	On

Sequence - Assistant

SAR Assistant	Off
Optimization	Min. TE

\\Research\Investigators\Frederick\Keto MRS 20240709\slicepositioning
TA: 20 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	70
Distance Factor	0 %
Position	R7.8 A14.8 F1.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	800.0 ms
TE	25.00 ms
Averages	1
Multi-band accel. factor	5
AutoAlign	Head > Brain

Contrast - Common

TR	800.0 ms
TE	25.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	52 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Delay in TR	0.00 ms

Resolution - Common

FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	108
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference scan mode	Single-shot
Acceleration Factor PE	2
Reference Lines PE	32
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Prescan

Geometry - Common

Slice Group	1
Slices	70
Distance Factor	0 %
Position	R7.8 A14.8 F1.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	5

Geometry - AutoAlign

Slice Group	1
Position	R7.8 A14.8 F1.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R7.8 A14.8 F1.0
R	7.8 mm
A	14.8 mm
F	1.0 mm
Initial Orientation	T > C
T > C	-20.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	1 mm
Table Position	F
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R7.8 A14.8 F1.0 mm
Orientation	T > C-20.0
Rotation	0.00 deg
A >> P	216 mm
R >> L	216 mm
F >> H	140 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	800.0 ms
Log Signals	Off
Multi-band accel. factor	5

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	1
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	2314 Hz/Px
Echo Spacing	0.54 ms
Free Echo Spacing	On
EPI Factor	108

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	4380 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	On
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	DICOM
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\Keto MRS 20240709\restingstate_mb5_AP
TA: 6:19 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	70
Distance Factor	0 %
Position	R7.8 A14.8 F1.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	800.0 ms
TE	25.00 ms
Averages	1
Multi-band accel. factor	5
AutoAlign	Head > Brain

Contrast - Common

TR	800.0 ms
TE	25.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	52 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	450
Delay in TR	0.00 ms

Resolution - Common

FOV Read	216 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	2.0 mm
Base Resolution	108
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference scan mode	Single-shot
Acceleration Factor PE	2
Reference Lines PE	32
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Prescan

Geometry - Common

Slice Group	1
Slices	70
Distance Factor	0 %
Position	R7.8 A14.8 F1.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	5

Geometry - AutoAlign

Slice Group	1
Position	R7.8 A14.8 F1.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R7.8 A14.8 F1.0
R	7.8 mm
A	14.8 mm
F	1.0 mm
Initial Orientation	T > C
T > C	-20.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	1 mm
Table Position	F
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R7.8 A14.8 F1.0 mm
Orientation	T > C-20.0
Rotation	0.00 deg
A >> P	216 mm
R >> L	216 mm
F >> H	140 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	800.0 ms
Log Signals	Off
Multi-band accel. factor	5

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	450
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	2314 Hz/Px
Echo Spacing	0.54 ms
Free Echo Spacing	Off
EPI Factor	108

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	4380 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	On
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	DICOM
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\Keto MRS 20240709\dMRI_dir64_MGH_AP

TA: 3:48 min Coil Selection: Auto Voxel Size: 1.5×1.5×1.5 mm^3 Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	92
Distance Factor	0 %
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.5 mm
TR	3230.0 ms
TE	89.20 ms
Multi-band accel. factor	4
AutoAlign	Head > Brain

Contrast - Common

TR	3230.0 ms
TE	89.20 ms
MTC	Off
Magn. Preparation	None
Flip Angle	78 deg
Refocus flip angle	160 deg
Fat-Water Contrast	Standard
Grad. rev. fat suppr.	Enabled
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.5 mm
Base Resolution	140
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	92
Distance Factor	0 %
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.5 mm
TR	3230.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	4

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H2.0
L	0.0 mm
P	0.0 mm
H	2.0 mm
Initial Orientation	T > C
T > C	-16.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	2 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	T > C-16.0
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	138 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3230.0 ms
Multi-band accel. factor	4

Physio - PACE

Resp. Control	Off

Physio - PACE

Multi-band accel. factor	4

Diff

Diffusion Mode	Free
Diff. Directions	64
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm²
b-value 2	2000 s/mm²
Averages 1	1
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	On
FA Maps	On
ADC Maps	On
Exponential ADC Maps	Off
ADC Noise Threshold	40
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	1700 Hz/Px
Echo Spacing	0.69 ms
Free Echo Spacing	Off
EPI Factor	140

Sequence - Part 2

Introduction	On
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	3840 us
Refocus pulse duration	7680 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	On
Invert RO/PE polarity	Off
PF omits higher k-space	Off
Force GPA balance	Off
Disable B1 control loop	Off

Sequence - Special

Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	DICOM

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\Keto MRS 20240709\fastestmap
TA: 6 sec Coil Selection: Manual Vol: 25×25×25 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
Vol R >> L	25 mm
Vol F >> H	25 mm
Vol A >> P	25 mm
TR	2000.0 ms
TE	49.20 ms
Averages	1
Coil Elements	HC3-6

Contrast - Common

TR	2000.0 ms
TE	49.20 ms
Tau	5.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Averages	1
Water Suppression	None

Resolution - Common

Vector Size	256
Normalize	Off

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
Vol R >> L	25 mm
Vol F >> H	25 mm
Vol A >> P	25 mm

Geometry - AutoAlign

AutoAlign	---

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	25 mm
F >> H	25 mm
A >> P	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2000.0 ms

Sequence - Common

Sequence Name	fastmp

Sequence - Common

Delta Frequency	0.00 ppm
Measurements	1
Phase Cycling	None
Save Uncombined	Off
Bandwidth	100000 Hz
Acquisition Duration	2 ms

Sequence - Special

Type of fit	Linear 3-proj
VoI fit factor	100.00 %
Force spherical fit VoI	Off Force spherical fit VoI
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.00 mm
Inversion pulse	Off
Multi-echo acquisition	On
Number of echoes	16

\\Research\Investigators\Frederick\Keto MRS 20240709\fastestmap
TA: 12 sec Coil Selection: Manual VoI: 25×25×25 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm
TR	2000.0 ms
TE	49.20 ms
Averages	1
Coil Elements	HC3-6

Contrast - Common

TR	2000.0 ms
TE	49.20 ms
Tau	5.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Averages	1
Water Suppression	None

Resolution - Common

Vector Size	256
Normalize	Off

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm

Geometry - AutoAlign

AutoAlign	---

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	25 mm
F >> H	25 mm
A >> P	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2000.0 ms

Sequence - Common

Sequence Name	fastmp

Sequence - Common

Delta Frequency	0.00 ppm
Measurements	1
Phase Cycling	None
Save Uncombined	Off
Bandwidth	100000 Hz
Acquisition Duration	2 ms

Sequence - Special

Type of fit	Full 6-proj
VoI fit factor	100.00 %
Force spherical fit VoI	Off Force spherical fit VoI
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.00 mm
Inversion pulse	Off
Multi-echo acquisition	On
Number of echoes	16

\\Research\Investigators\Frederick\Keto MRS 20240709\fastestmap
TA: 12 sec Coil Selection: Manual VoI: 25×25×25 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm
TR	2000.0 ms
TE	49.20 ms
Averages	1
Coil Elements	HC3-6

Contrast - Common

TR	2000.0 ms
TE	49.20 ms
Tau	5.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Averages	1
Water Suppression	None

Resolution - Common

Vector Size	256
Normalize	Off

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm

Geometry - AutoAlign

AutoAlign	---

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	25 mm
F >> H	25 mm
A >> P	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2000.0 ms

Sequence - Common

Sequence Name	fastmp

Sequence - Common

Delta Frequency	0.00 ppm
Measurements	1
Phase Cycling	None
Save Uncombined	Off
Bandwidth	100000 Hz
Acquisition Duration	2 ms

Sequence - Special

Type of fit	Full 6-proj
VoI fit factor	100.00 %
Force spherical fit VoI	Off Force spherical fit VoI
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.00 mm
Inversion pulse	Off
Multi-echo acquisition	On
Number of echoes	16

\\Research\Investigators\Frederick\Keto MRS 20240709\fastestmap
TA: 12 sec Coil Selection: Manual VoI: 25×25×25 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm
TR	2000.0 ms
TE	49.20 ms
Averages	1
Coil Elements	HC3-6

Contrast - Common

TR	2000.0 ms
TE	49.20 ms
Tau	5.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Averages	1
Water Suppression	None

Resolution - Common

Vector Size	256
Normalize	Off

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm

Geometry - AutoAlign

AutoAlign	---

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	25 mm
F >> H	25 mm
A >> P	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2000.0 ms

Sequence - Common

Sequence Name	fastmp

Sequence - Common

Delta Frequency	0.00 ppm
Measurements	1
Phase Cycling	None
Save Uncombined	Off
Bandwidth	100000 Hz
Acquisition Duration	2 ms

Sequence - Special

Type of fit	Full 6-proj
VoI fit factor	100.00 %
Force spherical fit VoI	Off Force spherical fit VoI
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.00 mm
Inversion pulse	Off
Multi-echo acquisition	On
Number of echoes	16

\\Research\Investigators\Frederick\Keto MRS 20240709\fastestmap
TA: 12 sec Coil Selection: Manual VoI: 25×25×25 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm
TR	2000.0 ms
TE	49.20 ms
Averages	1
Coil Elements	HC3-6

Contrast - Common

TR	2000.0 ms
TE	49.20 ms
Tau	5.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Averages	1
Water Suppression	None

Resolution - Common

Vector Size	256
Normalize	Off

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm

Geometry - AutoAlign

AutoAlign	---

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	25 mm
F >> H	25 mm
A >> P	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2000.0 ms

Sequence - Common

Sequence Name	fastmp

Sequence - Common

Delta Frequency	0.00 ppm
Measurements	1
Phase Cycling	None
Save Uncombined	Off
Bandwidth	100000 Hz
Acquisition Duration	2 ms

Sequence - Special

Type of fit	Full 6-proj
VoI fit factor	100.00 %
Force spherical fit VoI	Off Force spherical fit VoI
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.00 mm
Inversion pulse	Off
Multi-echo acquisition	On
Number of echoes	16

\\Research\Investigators\Frederick\Keto MRS 20240709\svs_slaser_dkd_WSFA
TA: 33 sec Coil Selection: Manual VoI: 25×25×25 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm
TR	3000.0 ms
TE 1	8.00 ms
TE 2	11.00 ms
TE 3	10.00 ms
Averages	1
Coil Elements	HC3-6

Contrast - Common

TR	3000.0 ms
TE 1	8.00 ms
TE 2	11.00 ms
TE 3	10.00 ms
Total TE	29.00 ms
Excitation Flip angle	90 deg
Refocusing Flip angle	180 deg
Preparation Scans	2
Averages	1
VAPOR WS	Enabled
Water Suppr. BW	70 Hz
VAPOR Flip angle	60.00 deg
Metabolite Cycling	Off

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	R4.4 A69.9 H7.7 mm

Geometry - Common

Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	25 mm
F >> H	25 mm
A >> P	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	slaser
Preparation Scans	2
Delta Frequency	0.00 ppm
Water Ref. Scan	Off
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	3000 Hz
Acquisition Duration	682 ms
Remove Oversampling	On

Sequence - Special

Excitation duration	2000.00 us
Refocusing duration	4000.00 us
AutoShim	Off
GOIA/FOCI	Off
Inversion pulse	Off
FA AutoCalib	Off
AutoVOI	Off
OVS module 1	On
OVS module 2	On
OVS module 3	On
OVS module 4	On
Extra OVS	Off
Advanced User	Off
OVS slab X	120.00 mm
OVS slab Y up	120.00 mm
OVS slab Y down	120.00 mm
OVS slab Z	120.00 mm
OVS gap	7.00 mm
OVS delay 7	83.00 ms
OVS delay 8	21.00 ms
Calibration Type	VAPOR WS FA
Debug Type	None
Increment step size	10 deg
Measurements	9
Spinal Cord Navigator	Off
Gradient Max. Amplitude	37 mT/m
Ramp time	200 us

\\Research\Investigators\Frederick\Keto MRS 20240709\svs_slaser_dkd
TA: 3:18 min Coil Selection: Manual Vol: 25×25×25 mm^3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
Vol R >> L	25 mm
Vol F >> H	25 mm
Vol A >> P	25 mm
TR	3000.0 ms
TE 1	8.00 ms
TE 2	11.00 ms
TE 3	10.00 ms
Averages	64
Coil Elements	HC3-6

Contrast - Common

TR	3000.0 ms
TE 1	8.00 ms
TE 2	11.00 ms
TE 3	10.00 ms
Total TE	29.00 ms
Excitation Flip angle	90 deg
Refocusing Flip angle	180 deg
Preparation Scans	2
Averages	64
VAPOR WS	Enabled
Water Suppr. BW	70 Hz
VAPOR Flip angle	110.00 deg
Metabolite Cycling	Off

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
Vol R >> L	25 mm
Vol F >> H	25 mm
Vol A >> P	25 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	25 mm
F >> H	25 mm
A >> P	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	slaser
Preparation Scans	2
Delta Frequency	0.00 ppm
Water Ref. Scan	Off
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	3000 Hz
Acquisition Duration	682 ms
Remove Oversampling	On

Sequence - Special

Excitation duration	2000.00 us
Refocusing duration	4000.00 us
AutoShim	Off
GOIA/FOCI	Off
Inversion pulse	Off
FA AutoCalib	Off
AutoVOI	Off
OVS module 1	On
OVS module 2	On
OVS module 3	On
OVS module 4	On
Extra OVS	Off
Advanced User	Off
OVS slab X	120.00 mm
OVS slab Y up	120.00 mm
OVS slab Y down	120.00 mm
OVS slab Z	120.00 mm
OVS gap	7.00 mm
OVS delay 7	80.00 ms
OVS delay 8	21.00 ms
Calibration Type	None
Debug Type	None
Spinal Cord Navigator	Off
Gradient Max. Amplitude	37 mT/m
Ramp time	200 us

\\Research\Investigators\Frederick\Keto MRS 20240709\svs_slaser_dkd_EC
TA: 30 sec Coil Selection: Manual VoI: 25×25×25 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm
TR	3000.0 ms
TE 1	8.00 ms
TE 2	11.00 ms
TE 3	10.00 ms
Averages	8
Coil Elements	HC3-6

Contrast - Common

TR	3000.0 ms
TE 1	8.00 ms
TE 2	11.00 ms
TE 3	10.00 ms
Total TE	29.00 ms
Excitation Flip angle	90 deg
Refocusing Flip angle	180 deg
Preparation Scans	2
Averages	8
VAPOR WS	RF off: Wref1
Water Suppr. BW	70 Hz
Metabolite Cycling	Off

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal

Geometry - Common

Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	25 mm
F >> H	25 mm
A >> P	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00

System - Tx/Rx

Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	slaser
Preparation Scans	2
Delta Frequency	0.00 ppm
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	3000 Hz
Acquisition Duration	682 ms
Remove Oversampling	On

Sequence - Special

Excitation duration	2000.00 us
Refocusing duration	4000.00 us
AutoShim	Off
GOIA/FOCI	Off
Inversion pulse	Off
FA AutoCalib	Off
AutoVOI	Off
OVS module 1	On
OVS module 2	On
OVS module 3	On
OVS module 4	On
Extra OVS	Off
Advanced User	Off
OVS slab X	120.00 mm
OVS slab Y up	120.00 mm
OVS slab Y down	120.00 mm
OVS slab Z	120.00 mm
OVS gap	7.00 mm
OVS delay 7	83.00 ms
OVS delay 8	21.00 ms
Calibration Type	None
Debug Type	None
Spinal Cord Navigator	Off
Gradient Max. Amplitude	37 mT/m
Ramp time	200 us

\\Research\Investigators\Frederick\Keto MRS 20240709\svs_slaser_dkd_wq
TA: 30 sec Coil Selection: Manual VoI: 25×25×25 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm
TR	3000.0 ms
TE 1	8.00 ms
TE 2	11.00 ms
TE 3	10.00 ms
Averages	8
Coil Elements	HC3-6

Contrast - Common

TR	3000.0 ms
TE 1	8.00 ms
TE 2	11.00 ms
TE 3	10.00 ms
Total TE	29.00 ms
Excitation Flip angle	90 deg
Refocusing Flip angle	180 deg
Preparation Scans	2
Averages	8
VAPOR WS	Off

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm

Geometry - Common

VoI F >> H	25 mm
VoI A >> P	25 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	25 mm
F >> H	25 mm
A >> P	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	slaser
Preparation Scans	2
Delta Frequency	0.00 ppm
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	3000 Hz
Acquisition Duration	682 ms
Remove Oversampling	On

Sequence - Special

Excitation duration	2000.00 us
Refocusing duration	4000.00 us
AutoShim	Off
GOIA/FOCI	Off
Inversion pulse	Off
FA AutoCalib	Off
AutoVOI	Off
Advanced User	Off
Calibration Type	None
Debug Type	None
Spinal Cord Navigator	Off
Gradient Max. Amplitude	37 mT/m
Ramp time	200 us

\\Research\Investigators\Frederick\Keto MRS 20240709\eja_svs_mpress_ws_rACC
TA: 41 sec Coil Selection: Manual VoI: 25×25×25 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm
TR	3000.0 ms
TE	68.00 ms
Averages	1
Coil Elements	HC3-6

Contrast - Common

TR	3000.0 ms
TE	68.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	2
Averages	1
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	110.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	1024
Normalize	Off

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	25 mm
F >> H	25 mm
A >> P	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	mpres
Preparation Scans	2
Delta Frequency	-1.70 ppm
Measurements	9
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	512 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	HC3-6

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	1.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1200 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Forced min. TE1	0 ms
MEGA flip angle	180 deg
VAPOR flip angle	100 deg
VAPOR delay 8	16 ms
VAPOR delay 7	60 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Use expt. SAR calc	Off
Weighted averages	Off
Check RF clipping	Off
Send ref. scans	Off
PRESS+4	Off
Symm. MEGA timing	Off

Sequence - Special

MEGA water suppr.	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	WS FA
WS FA inc.	10 deg
Measurements	9
Number of label freqs.	1
Editing pulse freq. [1]	1.90 ppm
Editing pulse BW	54.00 Hz

\\Research\Investigators\Frederick\Keto MRS 20240709\eja_svs_mpress_4p56Am25D1200bw54
TA: 13:14 min Coil Selection: Manual Vol: 25×25×25 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
Vol R >> L	25 mm
Vol F >> H	25 mm
Vol A >> P	25 mm
TR	3000.0 ms
TE	68.00 ms
Averages	128
Coil Elements	HC3-6

Contrast - Common

TR	3000.0 ms
TE	68.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	2
Averages	128
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	110.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	1024
Normalize	Off

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
Vol R >> L	25 mm
Vol F >> H	25 mm
Vol A >> P	25 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	25 mm
F >> H	25 mm
A >> P	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	mpres
Preparation Scans	2
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	512 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	HC3-6

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	25.00 mT/m
Spoiler amp. ratio	1.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1200 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	240 us
Forced min. TE1	0 ms
MEGA flip angle	180 deg
VAPOR flip angle	160 deg
VAPOR delay 8	16 ms
VAPOR delay 7	65 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	On
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
PRESS+4	Off
Symm. MEGA timing	Off

Sequence - Special

MEGA water suppr.	On
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	0.00 ppm
Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	54.00 Hz

\\Research\Investigators\Frederick\Keto MRS 20240709\eja_svs_mpress_rACC_EC
TA: 38 sec Coil Selection: Manual VoI: 25×25×25 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm
TR	3000.0 ms
TE	68.00 ms
Averages	4
Coil Elements	HC3-6

Contrast - Common

TR	3000.0 ms
TE	68.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	2
Averages	4
VAPOR	RF Off
VAPOR suppr.	Water Suppr.
Water s. BW	110.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	1024
Normalize	Off

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	25 mm
F >> H	25 mm
A >> P	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	mpres
Preparation Scans	2
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	512 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	HC3-6

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	25.00 mT/m
Spoiler amp. ratio	1.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1200 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	240 us
Forced min. TE1	0 ms
MEGA flip angle	180 deg
VAPOR flip angle	130 deg
VAPOR delay 8	16 ms
VAPOR delay 7	60 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	On
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
PRESS+4	Off
Symm. MEGA timing	Off

Sequence - Special

Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	7.50 ppm
Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	54.00 Hz

\\Research\Investigators\Frederick\Keto MRS 20240709\eja_svs_mpress_rACC_wq
TA: 38 sec Coil Selection: Manual VoI: 25×25×25 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm
TR	3000.0 ms
TE	68.00 ms
Averages	4
Coil Elements	HC3-6

Contrast - Common

TR	3000.0 ms
TE	68.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	2
Averages	4
VAPOR	None

Resolution - Common

Vector Size	1024
Normalize	Off

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm

Geometry - AutoAlign

AutoAlign	---

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	25 mm
F >> H	25 mm
A >> P	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	mpres

Sequence - Common

Preparation Scans	2
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	512 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	HC3-6

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	25.00 mT/m
Spoiler amp. ratio	1.00
Refocus grad. factor	1.00
OVS delta frequency	0.00 ppm
Spoiler duration	1200 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	240 us
Forced min. TE1	0 ms
MEGA flip angle	180 deg
Enable OVS	Off
Resolve averages	On
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
PRESS+4	Off
Symm. MEGA timing	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	0.00 ppm
Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	54.00 Hz

\\Research\Investigators\Frederick\Keto MRS 20240709\svs_se_lactate
TA: 13:00 min Coil Selection: Auto Vol: 25×25×25 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
Vol R >> L	25 mm
Vol F >> H	25 mm
Vol A >> P	25 mm
TR	3000.0 ms
TE	270.00 ms
Averages	256

Contrast - Common

TR	3000.0 ms
TE	270.00 ms
Flip Angle	90 deg
Preparation Scans	4
Averages	256
Water Suppression	Water Saturation
Water Suppr. BW	50 Hz
Spectral Suppr.	None

Resolution - Common

Vector Size	1024
Normalize	Off

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
Vol R >> L	25 mm
Vol F >> H	25 mm
Vol A >> P	25 mm

Geometry - AutoAlign

AutoAlign	---

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	On

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	25 mm
F >> H	25 mm
A >> P	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Physio - PACE

Resp. Control	Off

Sequence - Common

Sequence Name	svs_se
Preparation Scans	4
Delta Frequency	-3.30 ppm
Ref. Scan Mode	Off
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	512 ms
Remove Oversampling	On
Freq. Corr. Accumulation	Off
Incr. Gradient Spoiling	Off

\\Research\Investigators\Frederick\Keto MRS 20240709\svs_se_lactate_WS

TA: 1:00 min Coil Selection: Auto VoI: 25×25×25 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm
TR	3000.0 ms
TE	270.00 ms
Averages	16

Contrast - Common

TR	3000.0 ms
TE	270.00 ms
Flip Angle	90 deg
Preparation Scans	4
Averages	16
Water Suppression	RF Off
Water Suppr. BW	50 Hz
Spectral Suppr.	None

Resolution - Common

Vector Size	1024
Normalize	Off

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
VoI R >> L	25 mm
VoI F >> H	25 mm
VoI A >> P	25 mm

Geometry - AutoAlign

AutoAlign	---

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	On

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	25 mm
F >> H	25 mm
A >> P	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Physio - PACE

Resp. Control	Off

Sequence - Common

Sequence Name	svs_se
Preparation Scans	4
Delta Frequency	0.00 ppm
Ref. Scan Mode	Off
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	512 ms
Remove Oversampling	On
Freq. Corr. Accumulation	Off
Incr. Gradient Spoiling	Off